Signed Δ per photo for every model of each group — orange = predicted too early, blue = too late

(a) out-of-box encoders · 5 models
(b) SigLIP2 mean-pool: unfrozen blocks · 4 models
(c) DINOv3+BGE: unfrozen blocks · 5 models
(d) DINOv3+BGE: size and mining · 16 models

principal · fr
signed Δ (BBCH units)

too early (Δ < 0)
too late (Δ > 0)
chance IQR (25–75%)
chance median

SigLIP2-MP oob
CLIP-mp oob
CLIP-trunc oob
DINOv3 oob
random oob
SigLIP2-MP 0L
SigLIP2-MP 2L
SigLIP2-MP 4L
SigLIP2-MP 8L
DINOv3 0L
DINOv3 2L
DINOv3 4L
DINOv3 6L
DINOv3 8L
size 100
size 500
size 1000
size 2500
size 5000
size 10000
size 20000
size 33526
mined 100
mined 500
mined 1000
mined 2500
mined 5000
mined 10000
mined 20000
mined 33208

principal · en
signed Δ (BBCH units)

secondary · fr
signed Δ (BBCH units)

secondary · en
signed Δ (BBCH units)